

VOLOGDA, Russian Federation

WMO: 270370

Lat: 59.3201N

Lon: 39.9250E

Elev: 426

StdP: 14.47

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-14.9	-26.4	1.3	-20.7	-20.3	1.8	-14.8	21.3	20.9	18.9	23.1	4.3	320	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	83.4	67.8	79.6	65.9	76.2	64.1	70.0	79.8	67.9	76.7	65.9	73.8	7.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.0	74.6	64.5	92.4	72.3	62.6	86.4	70.4	34.3	80.0	32.5	77.1	30.9	73.9	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.7	14.6	DB	-27.1	87.4	6.9	3.6	-32.0	90.1	-36.0	92.2	-39.9	94.2	-44.9	96.9	(4)
(5)			WB	-27.3	72.8	6.8	3.2	-32.2	75.1	-36.2	77.0	-40.0	78.8	-44.9	81.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	13.1	14.4	24.3	39.1	51.6	58.9	64.2	59.5	49.9	38.5	26.6	18.4		
	DBStd	20.38	14.44	14.27	9.65	8.70	8.95	7.47	6.39	6.56	6.88	7.76	10.72	13.59		
	HDD50	5518	1144	996	797	343	91	10	1	5	84	364	703	980		
	HDD65	9859	1609	1416	1262	776	423	212	93	195	453	822	1153	1445		
	CDD50	1262	0	0	0	17	141	277	440	299	81	7	0	0		
	CDD65	128	0	0	0	0	8	28	68	23	1	0	0	0		
	CDH74	1235	0	0	0	5	87	312	613	213	5	0	0	0		
	CDH80	286	0	0	0	0	12	72	153	49	0	0	0	0		
	WSAvg	7.2	7.3	7.5	7.7	7.3	8.0	6.9	6.1	6.3	6.7	7.4	7.5	7.6		
	PrecAvg	22.1	1.3	1.0	1.0	1.3	1.7	2.5	2.9	2.8	2.2	2.0	1.7	1.6		
(15) Precipitation	PrecMax	28.0	2.6	2.2	2.7	2.7	3.8	7.1	6.5	6.9	4.1	3.7	3.2	2.8		
	PrecMin	14.2	0.2	0.0	0.1	0.1	0.5	0.4	0.4	0.0	0.3	0.0	0.3	0.6		
	PrecStd	2.8	0.5	0.5	0.6	0.7	0.9	1.3	1.6	1.4	0.9	0.9	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.9	37.5	49.4	72.6	82.0	86.2	88.4	85.6	74.3	62.9	47.1	41.9		
		MCWB	34.3	34.8	40.8	55.7	64.2	68.9	70.7	71.3	62.1	53.7	45.5	40.7		
	2%	DB	34.2	35.3	42.7	64.3	76.1	82.0	84.2	79.2	69.0	55.4	43.4	38.4		
		MCWB	33.0	33.4	36.6	50.4	60.2	66.6	68.7	65.9	60.0	50.9	41.8	37.1		
	5%	DB	33.3	33.9	39.0	58.9	71.7	77.3	80.5	75.3	64.6	51.9	41.1	35.2		
		MCWB	32.4	32.5	34.7	47.7	57.4	63.9	67.2	64.4	57.6	48.7	39.8	33.9		
	10%	DB	31.8	32.4	36.5	53.3	67.7	73.3	76.9	71.7	61.0	49.3	38.5	33.6		
		MCWB	31.0	31.2	33.4	44.4	54.8	62.1	65.8	62.4	55.8	46.9	37.5	32.6		
	0.4%	WB	34.7	35.5	41.6	55.8	67.1	71.4	73.1	72.3	64.3	55.7	46.0	41.0		
		MCDB	35.3	36.5	47.7	71.3	77.3	82.1	84.0	82.8	71.6	60.7	46.4	41.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.3	33.8	37.7	51.5	62.5	68.6	70.6	67.9	61.0	51.9	42.1	37.1		
		MCDB	33.9	34.7	40.8	62.4	72.6	78.7	80.5	76.1	66.4	54.6	43.0	38.1		
	5%	WB	32.4	32.6	35.5	48.5	59.5	65.9	68.7	65.6	58.7	49.3	39.8	34.1		
		MCDB	33.2	33.7	37.9	56.3	68.1	74.7	77.7	72.8	63.1	51.5	40.8	34.8		
	10%	WB	30.7	31.2	33.9	45.3	56.6	63.1	66.9	63.6	56.3	46.8	37.4	32.5		
		MCDB	31.6	32.5	35.9	52.4	65.3	70.8	75.1	70.2	60.7	49.0	38.4	33.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	13.3	16.3	17.6	19.6	18.8	18.9	17.6	14.7	9.6	7.7	9.4		
		MCDBR	8.3	9.5	17.6	25.8	25.8	24.8	24.1	23.4	20.4	14.8	8.7	6.3		
		MCWBR	8.3	8.8	12.6	14.5	12.4	11.9	10.9	11.5	11.8	10.8	8.1	6.3		
		MCDBR	8.0	8.7	13.3	23.6	21.5	22.2	21.4	20.5	17.5	13.7	8.1	6.4		
	5% WB	MCWBR	8.0	8.4	10.0	14.2	11.6	11.7	10.6	11.1	10.9	10.5	8.0	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.308	0.334	0.381	0.362	0.360	0.393	0.388	0.353	0.332	0.294	0.251		
	taud		2.061	2.063	2.089	2.185	2.407	2.430	2.357	2.349	2.430	2.366	2.164	2.026		
	Ebn at Noon		160	213	248	259	275	277	263	255	246	212	150	121		
	Edh at Noon		16	27	35	38	33	33	35	33	26	20	14	11		
(44) All-Sky Solar Radiation	RadAvg		114	342	786	1229	1645	1734	1704	1289	745	321	105	61		
	RadStd		18	71	102	141	106	116	192	86	82	52	18	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+1.24	N/A	N/A	N/A	N/A	N/A	-322	-396	N/A	N/A

Nomenclature: See separate page